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1. Introduction

The purpose of this Tier 2 SOC lab is to simulate investigation and response activities beyond Tier 1 monitoring. This involves setting up Sysmon to collect advanced endpoint telemetry, configuring data collection into Azure Sentinel, and using queries to investigate potential security incidents.

2. Objectives

- Install and configure Sysmon to generate detailed endpoint logs.
- Configure Data Collection Rule (DCR) to ingest Sysmon events into Log Analytics.
- Build advanced queries (KQL) for investigation.
- Demonstrate containment steps (account isolation, host isolation, etc.).
- Practice **false-positive tuning** and basic automation.

3. Overview

In Tier 1, I successfully:

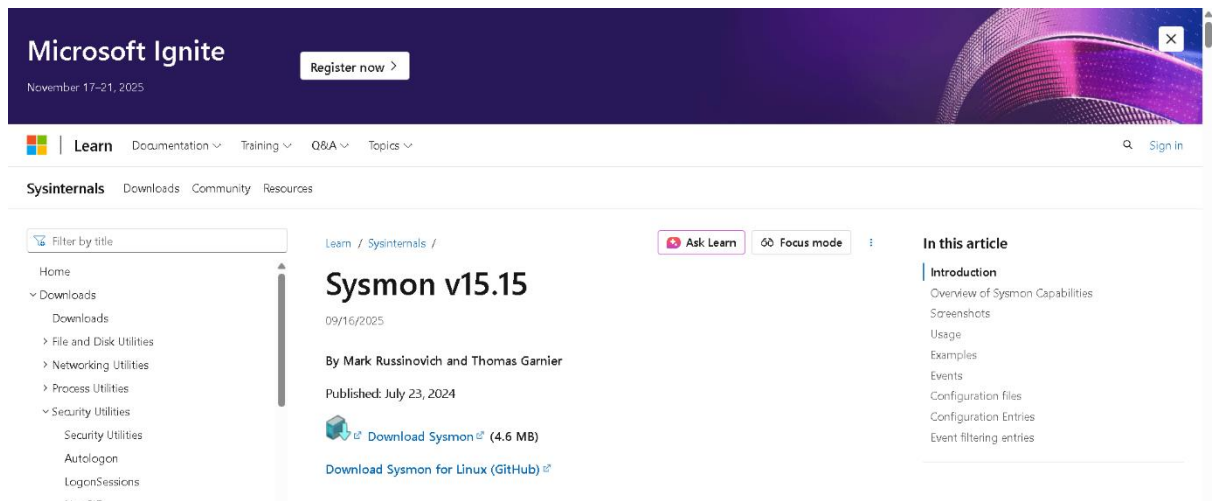
- Created a Log Analytics workspace.
- Connected it to Microsoft Sentinel.
- Configured AMA and baseline DCR.
- Collected and queried Security Event logs (failed/successful logons).

In Tier 2, the focus is on deep system monitoring via Sysmon logs, advanced KQL queries, and response actions.

4. Step-by-Step Explanation

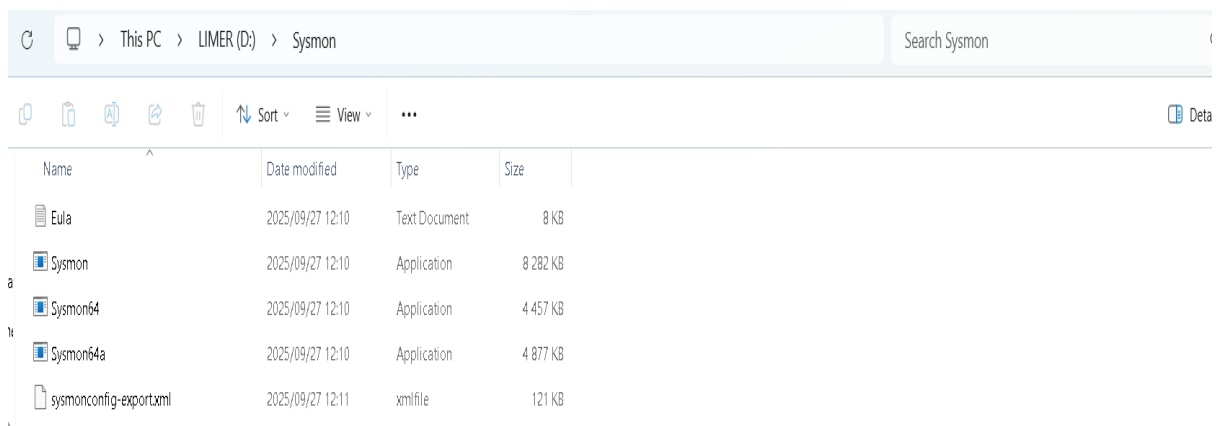
a. Downloaded Sysmon

- From Microsoft Sysinternals:



b. Extracted Files

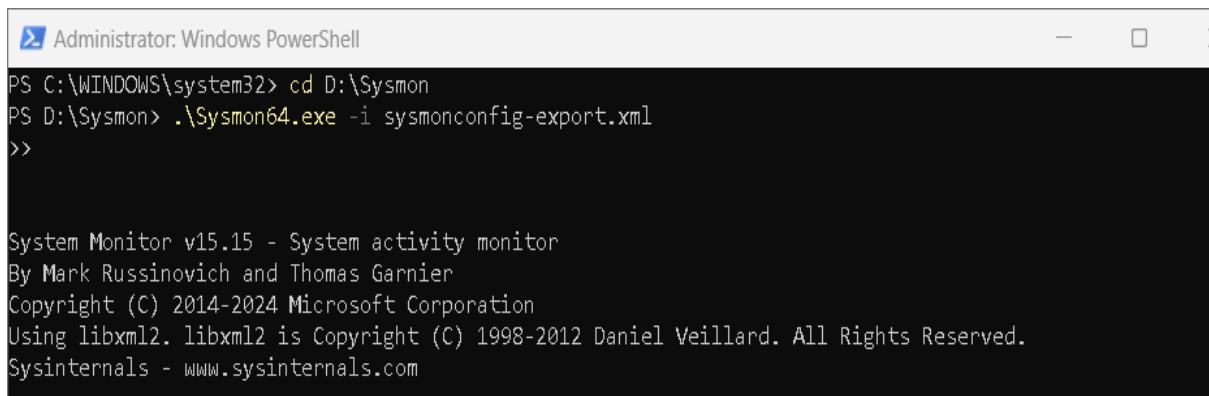
- Extracted all files to D:\Sysmon



c. Downloaded a Sysmon Config File

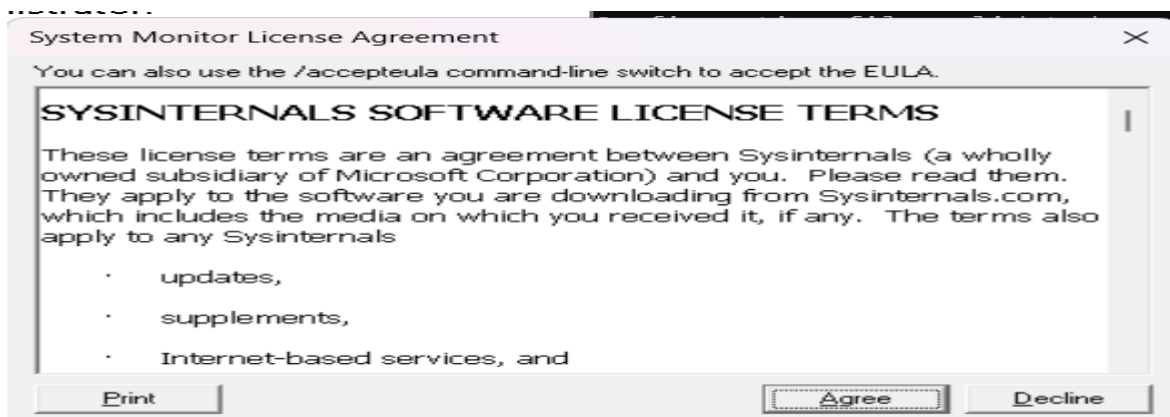
- Used sysmonconfig-export.xml to define event collection.
- Saved config in the same folder:

D:\Sysmon\sysmonconfig-export.xml



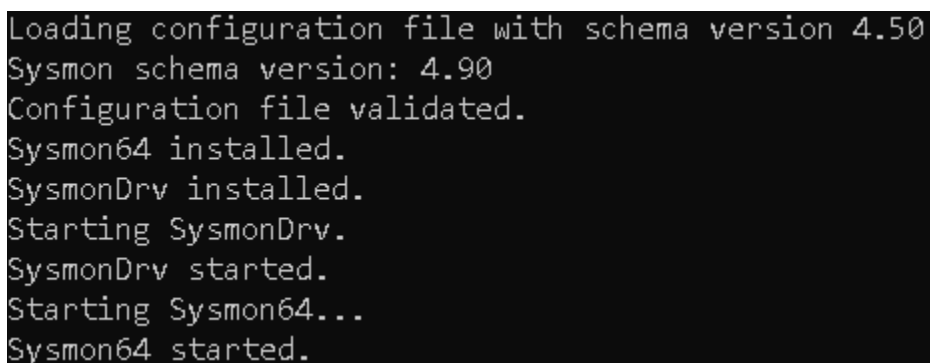
```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> cd D:\Sysmon
PS D:\Sysmon> .\Sysmon64.exe -i sysmonconfig-export.xml
>>

System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com
```



d. Installed Sysmon with Config

- Opened PowerShell (Administrator).
- Ran:
 `.\Sysmon64.exe -i sysmonconfig-export.xml`



```
Loading configuration file with schema version 4.50
Sysmon schema version: 4.90
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64...
Sysmon64 started.
```

e. Verified Sysmon is Running

- Checked service status:

Get-Service sysmon64

```
PS D:\Sysmon> Get-Service sysmon64
>>

Status      Name                DisplayName
-----
Running     Sysmon64            sysmon64

PS D:\Sysmon>
```

f. Configured DCR for Sysmon

- Went to DCR-AMA in Azure.

- Added Windows Event Logs source:

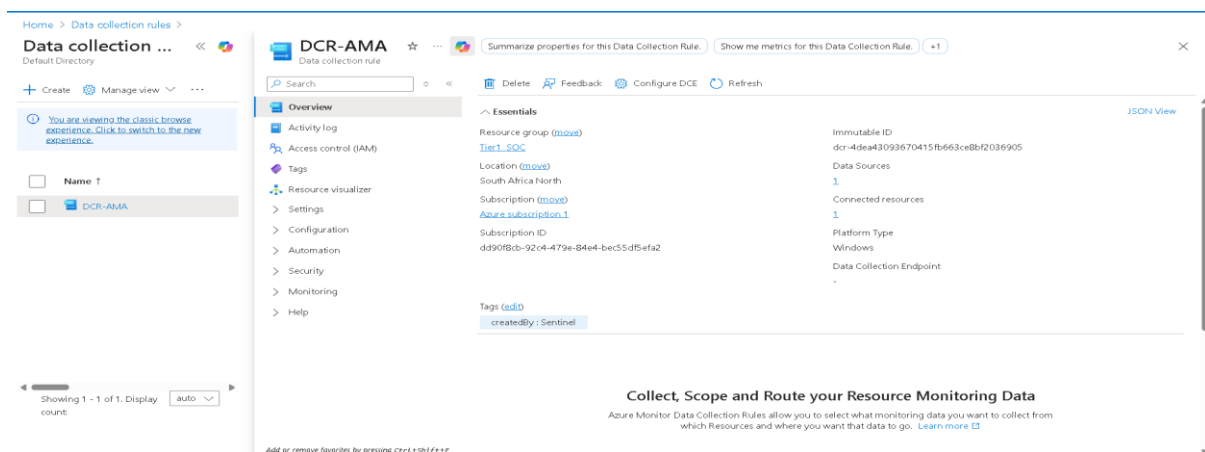
Microsoft-Windows-Sysmon/Operational

Destination: Log Analytics workspace (connected to Sentinel).

Event

| where LogName == "Microsoft-Windows-Sysmon/Operational"

| take 20



5. Findings / Analysis

- SecurityEvent Logs (Tier 1): Showed failed/successful logons.
- Sysmon Logs (Tier 2): Captured detailed process creation, network connections, and driver loads.
- This allows building timelines and investigating attacks at a granular level.

a. Evidence

- Example Queries
- Process Creation Events:

Event

```
| where LogName == "Microsoft-Windows-Sysmon/Operational"
```

```
| where EventID == 1
```

```
| project TimeGenerated, Computer,  
tostring(EventData.CommandLine), tostring(EventData.Image)
```

```
| sort by TimeGenerated desc
```

Network Connections (Outbound):

Event

```
| where LogName == "Microsoft-Windows-Sysmon/Operational"
```

```
| where EventID == 3
```

```
| project TimeGenerated, Computer,  
tostring(EventData.DestinationIp),  
tostring(EventData.DestinationPort)
```

6. Actions Taken

- Installed Sysmon and validated event capture.
- Configured DCR to collect Sysmon logs.
- Queried Sysmon logs in Sentinel.
- Prepared enrichment plans (VirusTotal, OTX, GreyNoise).
- Simulated investigation workflow.

7. Lessons Learned & Prevention

- Sysmon logs are richer than default Security logs and critical for deeper analysis.
- DCR setup must be log-based, not metrics-only; otherwise, Sysmon cannot be collected.
- Queries must map correctly to the event schema (e.g., EventID 1 = Process Creation, EventID 3 = Network Connection).
- Next step: Automate enrichment with SOAR playbooks.

8. Difficulties

Sysmon Event Log Path Issue

1. I struggled to find the correct location for Microsoft-Windows-Sysmon/Operational under the event log settings.
2. The log channel was not visible by default, which made it difficult to configure queries in Sentinel.
3. This caused delays in confirming whether Sysmon was correctly installed and generating events.

DCR-AMA (Data Collection Rule – Azure Monitor Agent) Issue

1. When setting up data collection, certain options in the DCR (Data Collection Rule) interface were greyed out / disabled.
2. This limited my ability to select specific Sysmon logs for ingestion into Sentinel.
3. The issue seemed tied to permissions or policy restrictions, preventing me from fully customizing what data could be collected.

Add data source ×

* Data source

Destination

Select which data source type and the data to collect for your resource(s).

Data source type *

Select a data type

Performance Counters

Windows Event Logs

IIS Logs

Firewall Logs (No data collection endpoint selected)

Custom Text Logs

Custom JSON Logs

Prometheus Metrics (No data collection endpoint selected)

Not yet done:

1. Custom Analytics Rule (full): Still need to create a saved Analytics Rule in Sentinel with severity, suppression, and entity mapping.
2. False-Positive Tuning: No before/after demonstration yet (e.g., reducing noise from repeated failed logons).
3. SOAR Enrichment (OTX, VT, GreyNoise): No playbook setup yet for auto-enrichment.
4. Containment Drill: No test done for disabling an account, revoking sessions, or resetting a password.
5. Threat Intelligence Watchlist: No IOC CSV uploaded and correlated yet.
6. Post-Incident Review (formal): You drafted lessons learned, but not a full “gap detection ticket.”

9. Conclusion

This Tier 2 lab extended Tier 1 monitoring by adding advanced endpoint telemetry (Sysmon), configuring proper data ingestion into Sentinel, and building investigation queries. This provides a stronger foundation for incident response, containment, and threat hunting.